

Binghan Xue

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EDUCATION

University of Maryland, College Park

Ph.D. in Neuroscience (GPA 3.5/4.0)

College Park, MD

August 2017 – May 2023

China Agricultural University

B.Sc. in Agronomy (GPA 3.8/4.0), B.B.A in Business Administration (GPA 3.6/4.0)

Beijing, China

August 2013 – July 2017

SKILLS

- Programming Language: Python; SQL; Matlab; Palantir; Cloud Computing (AWS); R; JavaScript; CSS; HTML
- Deep Learning (CNN, LSTM, etc); Natural Language Processing (BERT); Reinforcement Learning; API Development; Web development

PROFESSIONAL EXPERIENCE

National Center for Advancing Translational Sciences, NIH

Developer

Bethesda, MD

July 2023 – Present

- Designed and developed the whole data lake pipeline in Palantir, encompassing tasks such as cleaning, transforming, integrating, and harmonizing multiple data sources, while also executing rigorous QA/QC, UAT, and Production tests tailored specifically for rare disease data
- Utilized AI tools to predict drug-associated gene targets from thousands of Tox21 screening data, unveiling a discovery of 220 novel drug-gene pairs which hold immense promise for drug repurposing of rare diseases
- Co-led the drug repurposing team, in collaboration with the National Human Genome Research Institute, to develop advanced deep learning models aimed at identifying novel therapeutic opportunities for rare diseases with AoU data

Johns Hopkins University

Research Assistant

Baltimore, MD

August 2019 – May 2023

- Combined clustering (K-Means) with dimensional reduction methods (PCA) to redefine the subtype of neurons from more than 4000 neurons, classifying neurons into 6 different groups including 4 novel subtypes
- Proposed predictive models for brain circuits reconstruction based on experiments and wrote Matlab script to simulate the model, validating existence theory from a new perspective and substantiated experiment results

University of Maryland, College Park

Research Assistant

College Park, MD

August 2017 – August 2019

- Designed a state-of-the-art data processing platform (a Graphic User interface) for effectively analyzing large volume of electrophysiological data including data cleaning, signal processing, event capturing, regression fitting, dimensionality reducing, and statistical analysis and the platform has been widely used in research projects
- Built efficient image analysis workflows with auto encoder to provide rigorous, quantitative description of imaging time-series datasets and extract informative signal data for further analysis
- Established a distinctive framework with Python script for cortical microcircuits analysis by introducing 2-D Fast Fourier Transform (FFT), which unraveled the spatial distribution of neuron connections

SELECTED CONFERENCE PRESENTATION

" Prediction of Drug Targets Based on In Vitro Activity Profiles Toward Drug Repurposing for Rare Diseases"
International Conference on Bioinformatics and Biomedicine, Istanbul, Turkey, December, 2023

" Sex-specific age-related changes in excitatory and inhibitory intra-cortical circuits in mouse primary auditory cortex"
Association for research in Otolaryngology, February 2021

Award

Director's Award

Dean's Fellowship

National Center for Advancing Translational Sciences (2024)

University of Maryland (2017)

PUBLICATION

- Xue, B., Huang, R., Zhu, Q., & Xu, Y. (2023, December). Novel Target Identification towards Drug Repurposing Based on Biological Activity Profiles. (Under PLOS Biology review)
- Xue, B., Huang, R., Zhu, Q., & Xu, Y. (2023, December). Prediction of Drug Targets based on In Vitro Activity Profiles Toward Drug Repurposing for Rare Diseases. In 2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM) (pp. 2761-2763). IEEE.
- Xue, B., Meng, X., Kao, J. P., & Kanold, P. O. (2023). Age-related changes in excitatory and inhibitory intra-cortical circuits in auditory cortex of C57Bl/6 mice. *Hearing research*, 429, 108685.
- Xue, B., Alipio, J. B., Kao, J. P., & Kanold, P. O. (2022). Perinatal Opioid Exposure Results in Persistent Hypoconnectivity of Excitatory Circuits and Reduced Activity Correlations in Mouse Primary Auditory Cortex. *Journal of Neuroscience*, 42(17), 3676-3687
- Xue, B., Meng, X., Xu, Y., Kao, J. P., & Kanold, P. O. (2022). Transient coupling between infragranular and subplate layers to Layer 1 neurons before ear opening and throughout the critical period depends on peripheral activity. *Journal of Neuroscience*, 42(9), 1702-1718
- Xue, B., Kao, J. P., & Kanold, P. O. (2022). Sex-specific age-related changes in excitatory and inhibitory intra-cortical circuits in mouse primary auditory cortex. *bioRxiv*, 2022-06.
- Mukherjee, D., Xue, B., Chen, C. T., Chang, M., Kao, J. P., & Kanold, P. O. (2023). Early retinal deprivation crossmodally alters nascent subplate circuits and activity in the auditory cortex during the precritical period. *Cerebral Cortex*, 33(14), 9038-9053.